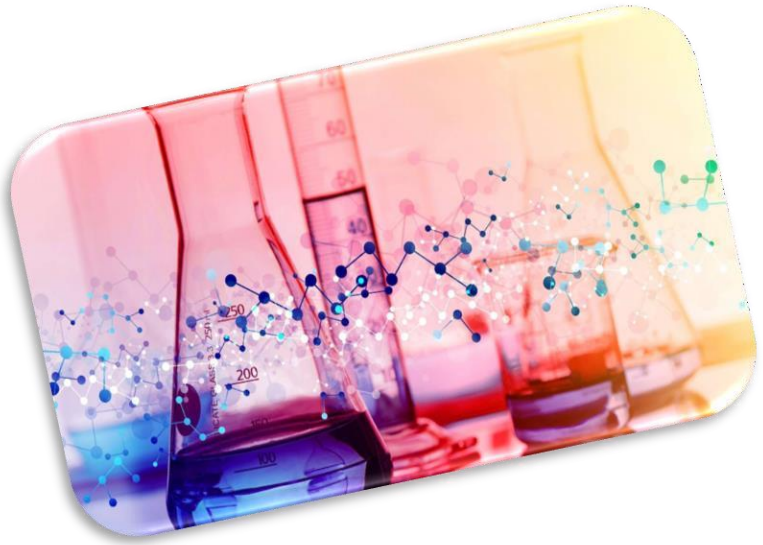
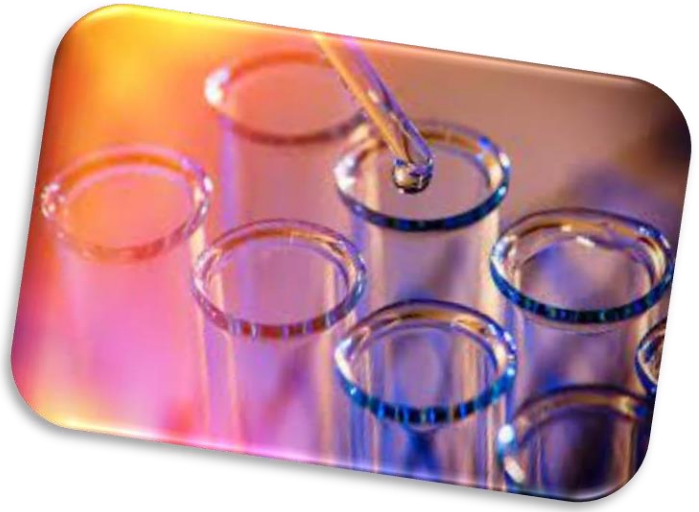
Arab Academy
for Science, Technology & Maritime Transport

Biochemistry
EBA1300





Dr. Mohamed R. Eletmany

PhD in Fiber Science and Polymer Chemistry

NC State Wilson College of Textiles, NC State University, North Carolina, USA

Lecture 5

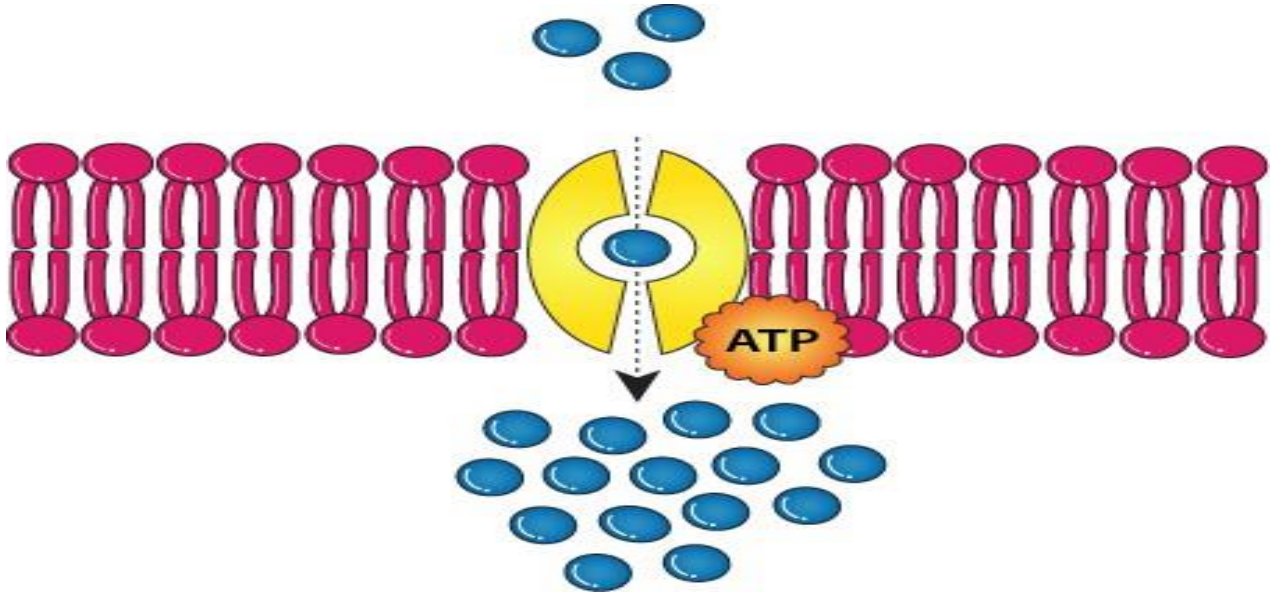
Cell Transport
(Active & Bulk Transport)

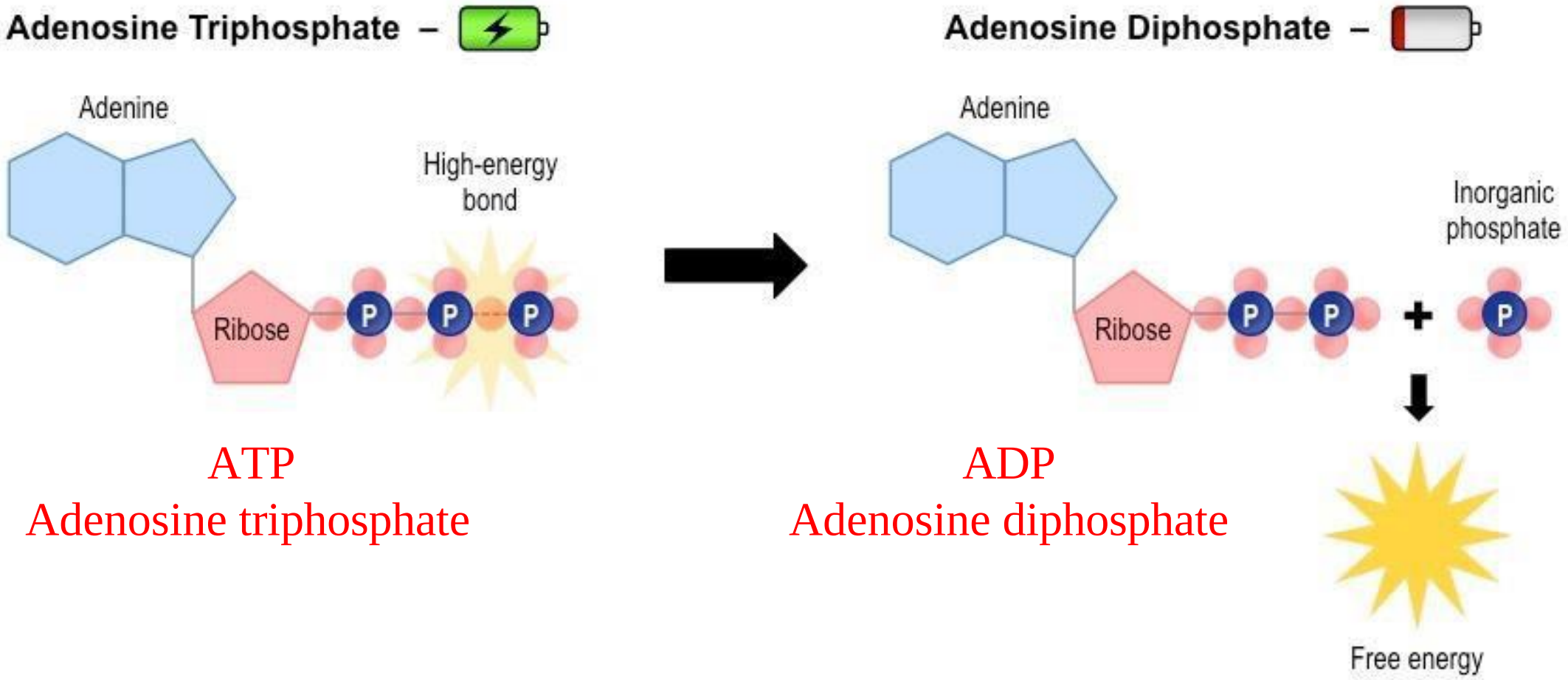
Important Terms

- **Concentration**: depends on the amount of solute in a solution (high concentration means high amount of solute).
- **Solute**: the dissolved substance in a solution (ex. sugar or salt).
- **Solvent**: is the substance that dissolves other substances (solutes) to make a solution (ex. water).
- **Solution**: a mixture of solvent and solute (ex. water + salt).
- **Concentration gradient**: is the difference in the concentration of solutes across a membrane.
- **Electrochmical gradient**: is the difference in the concentration of ions across a membrane.

Active transport

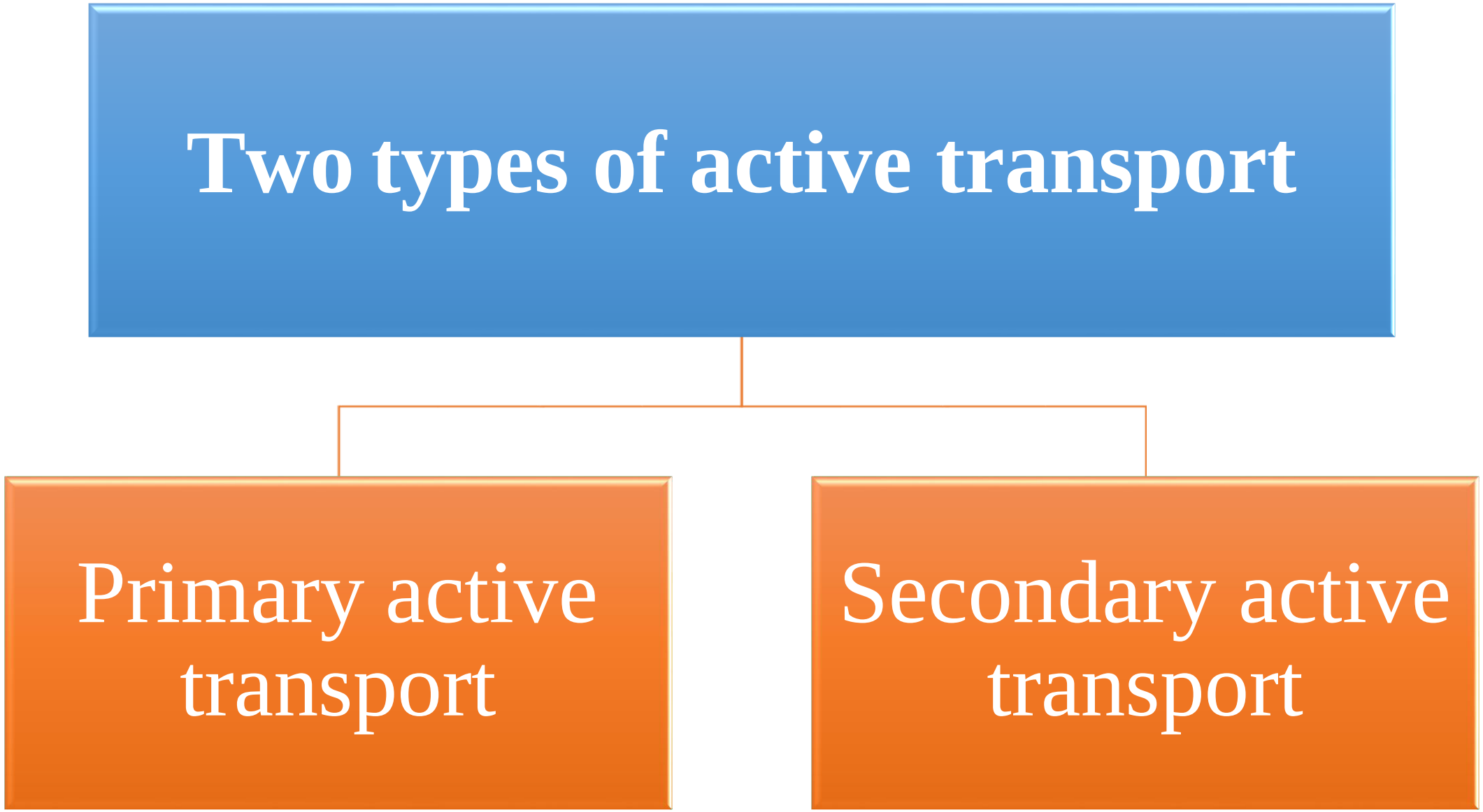
- It is the movement of substances (molecules) from **Lower** to **higher** concentration.
- This means that the movement of molecules is **against their concentration gradient**.
- So, energy (ex. ATP) is needed to pump the molecules against the concentration gradient.
- Usually, proteins are needed to transport molecules are called **pump proteins or transporters**.





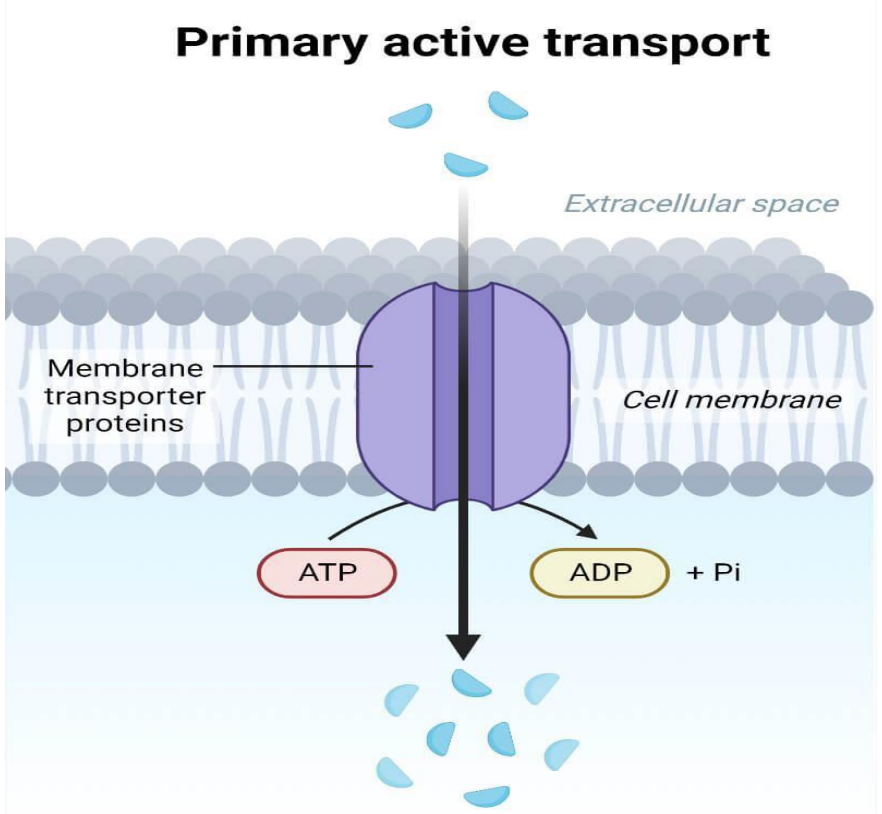
ATP
Adenosine triphosphate

ADP
Adenosine diphosphate


Two types of active transport
Primary active transport
Secondary active transport

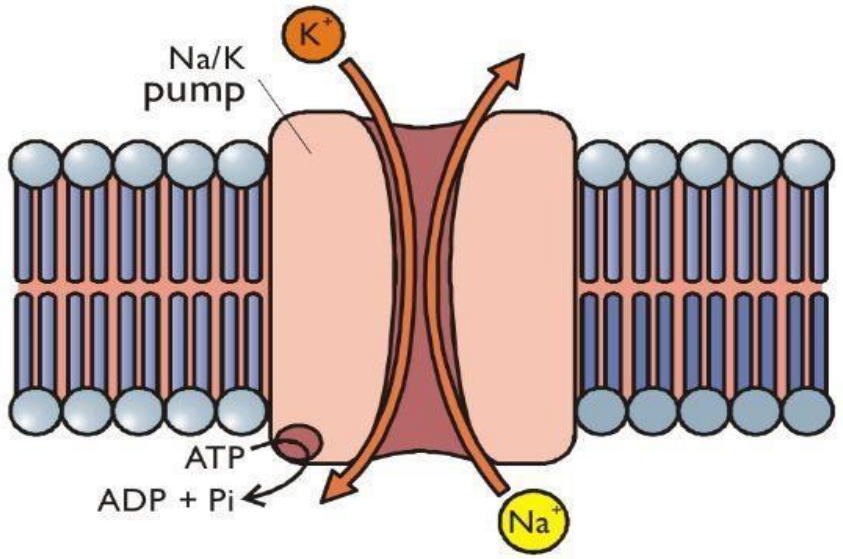
Primary active transport

- It is called Direct Active Transport.
- Uses ATP as a source of energy (Energy comes from hydrolysis of ATP).



Primary active transport

- Molecules transported by primary active transport are metal ions, such as Na^+, K^+, Mg^{2+}, and Ca^{2+}
- These ions require ion pumps to move (ex. Sodium-potassium pump and calcium pump).
- These pumps have binding site for ATP.



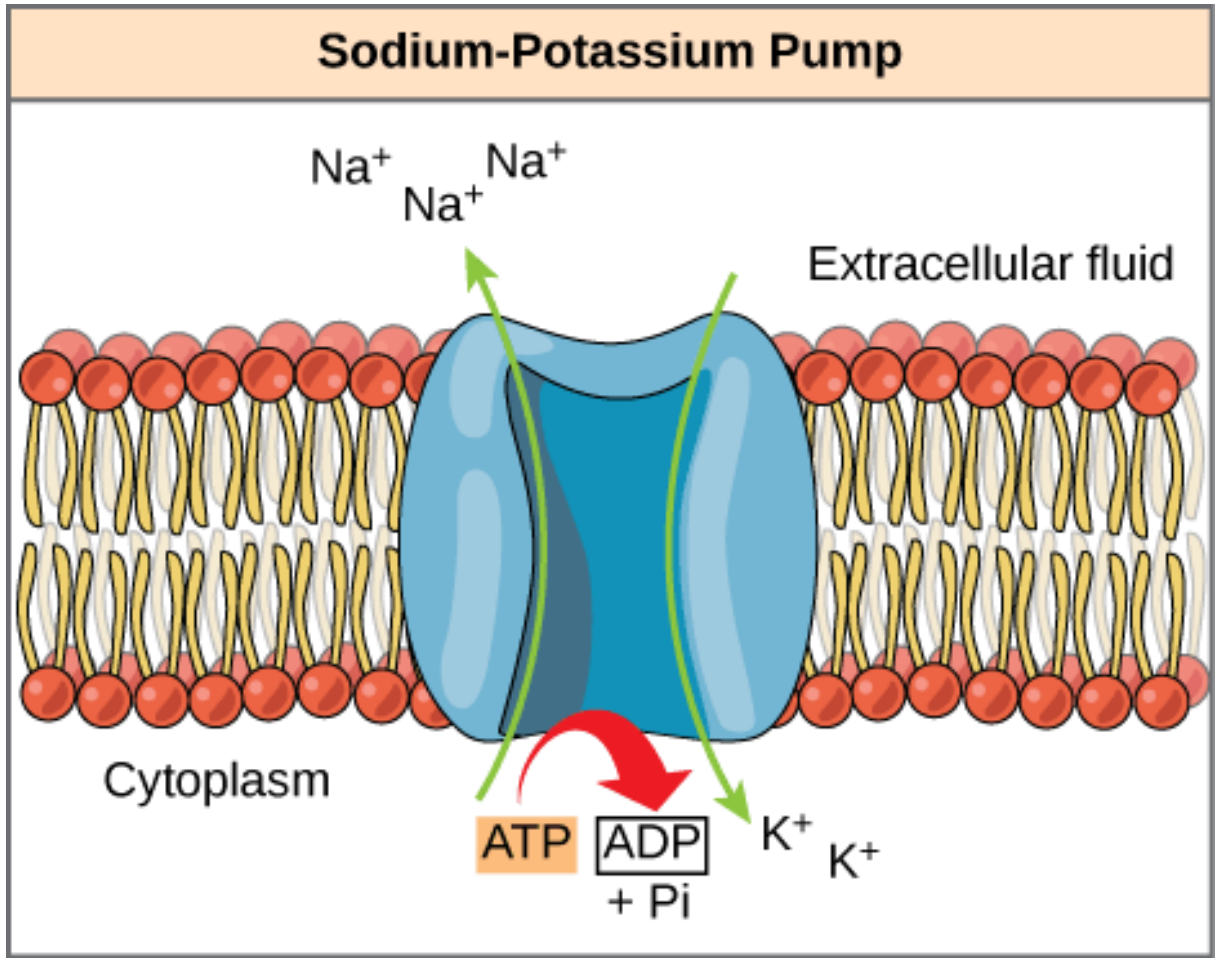
Sodium-Potassium pump

- All eukaryotes have sodium-potassium pump in their cells.

- **<u>This Na/ K pump (protein) is important to:</u>**

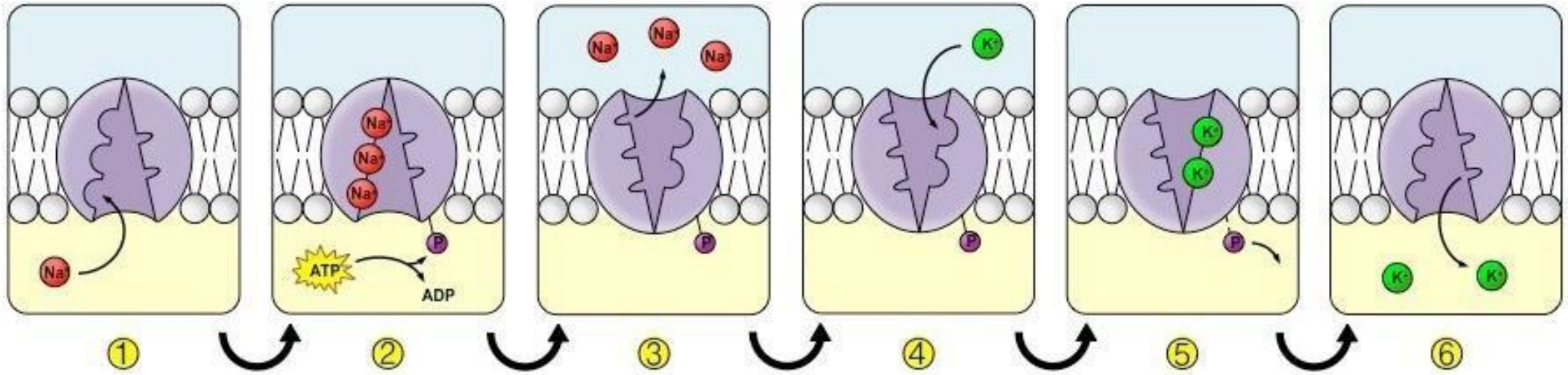
- Maintain correct concentrations of Na^+ and K^+ in living cells.
- Maintain cell volume.
- Maintain membrane voltage, so known as (**electrogenic pump).**

- The sodium-potassium pump transports sodium out of the cell and potassium into the cell in a repeating cycle of conformational (shape) changes of the pump.
- In each cycle, three sodium ions exit the cell, while two potassium ions enter. This process takes place in the following steps:


Sodium-Potassium Pump
Na+
Na+
Na+
Extracellular fluid
Cytoplasm
ATP
ADP
+ Pi
K+
K+

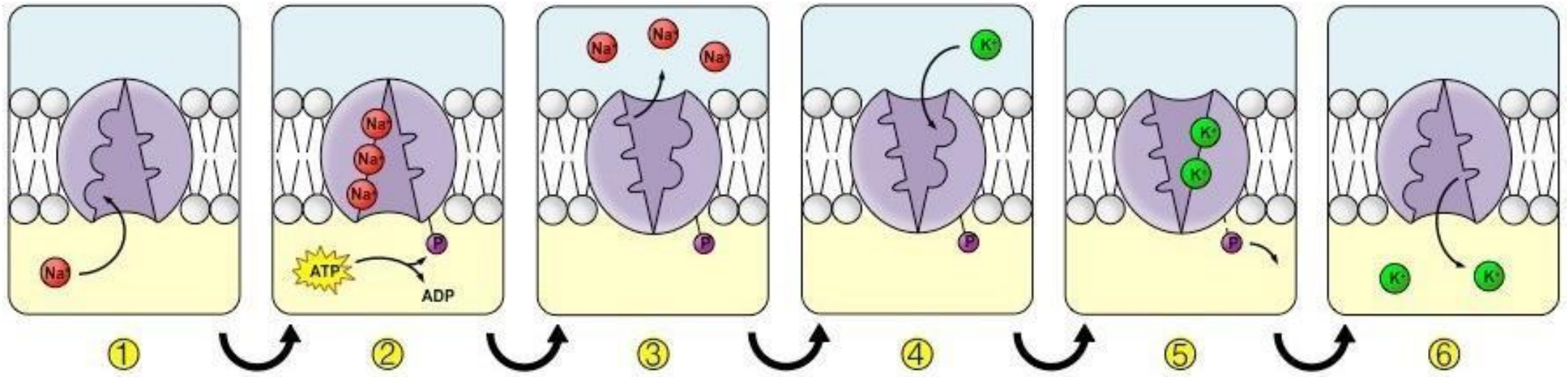
Sodium-Potassium pump

- To begin, the pump opens to the inside of the cell, then it binds to three sodium ions.
- When the three sodium ions bind, this causes hydrolysis (break down) of ATP to phosphate and ADP.



Sodium-Potassium pump

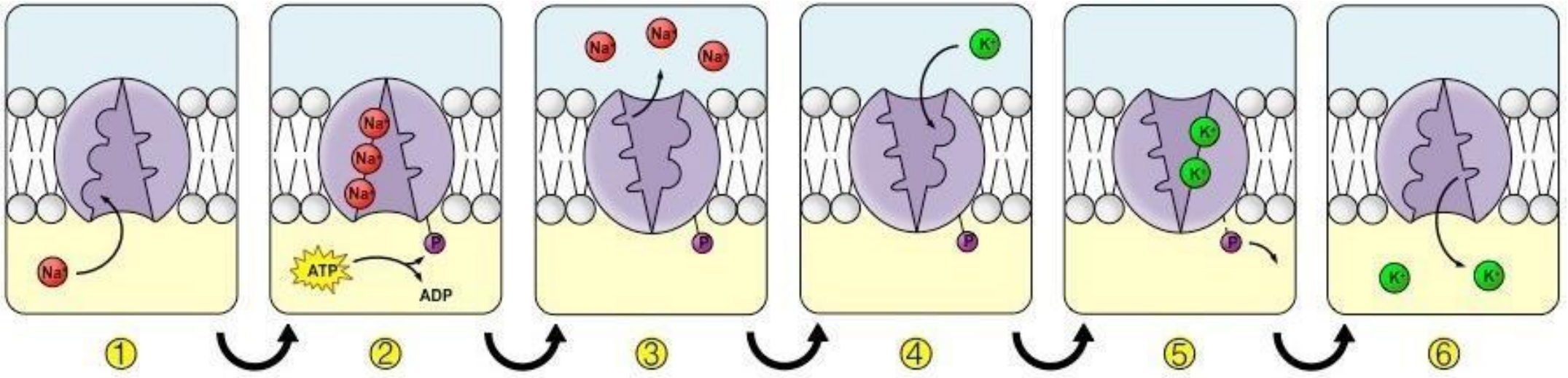
- The phosphate is attached to the pump, and the pump is then said to be (phosphorylated).
- Phosphorylation makes the pump to change its shape, making it to open towards the extracellular space and release three sodium ions outside the cell.



Sodium-Potassium pump

- After that, this pump binds to two potassium ions, and this causes removal (release) of the phosphate group that attached to it (dephosphorylated).

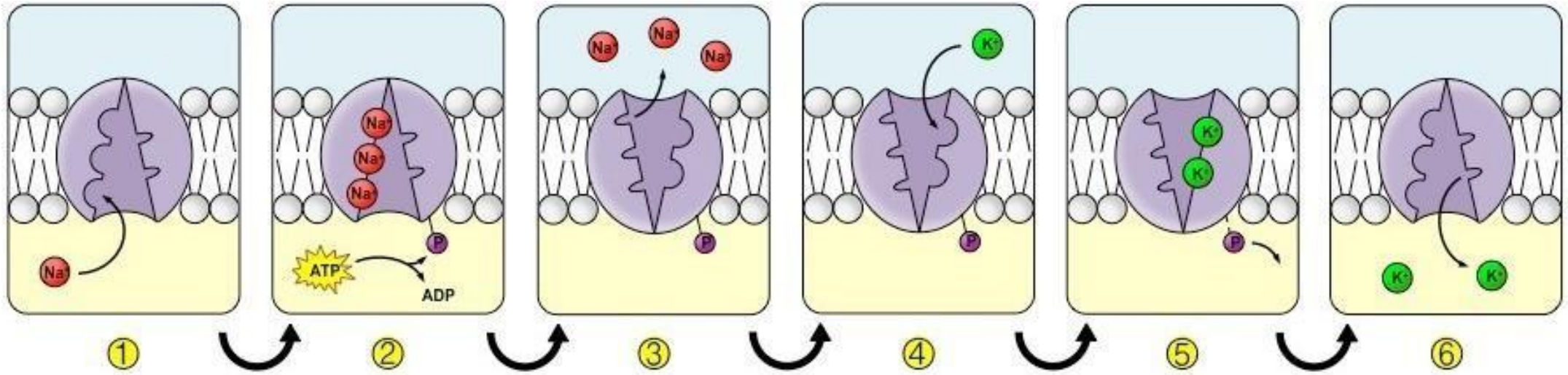
- Then, the pump will open towards the interior of the cell.

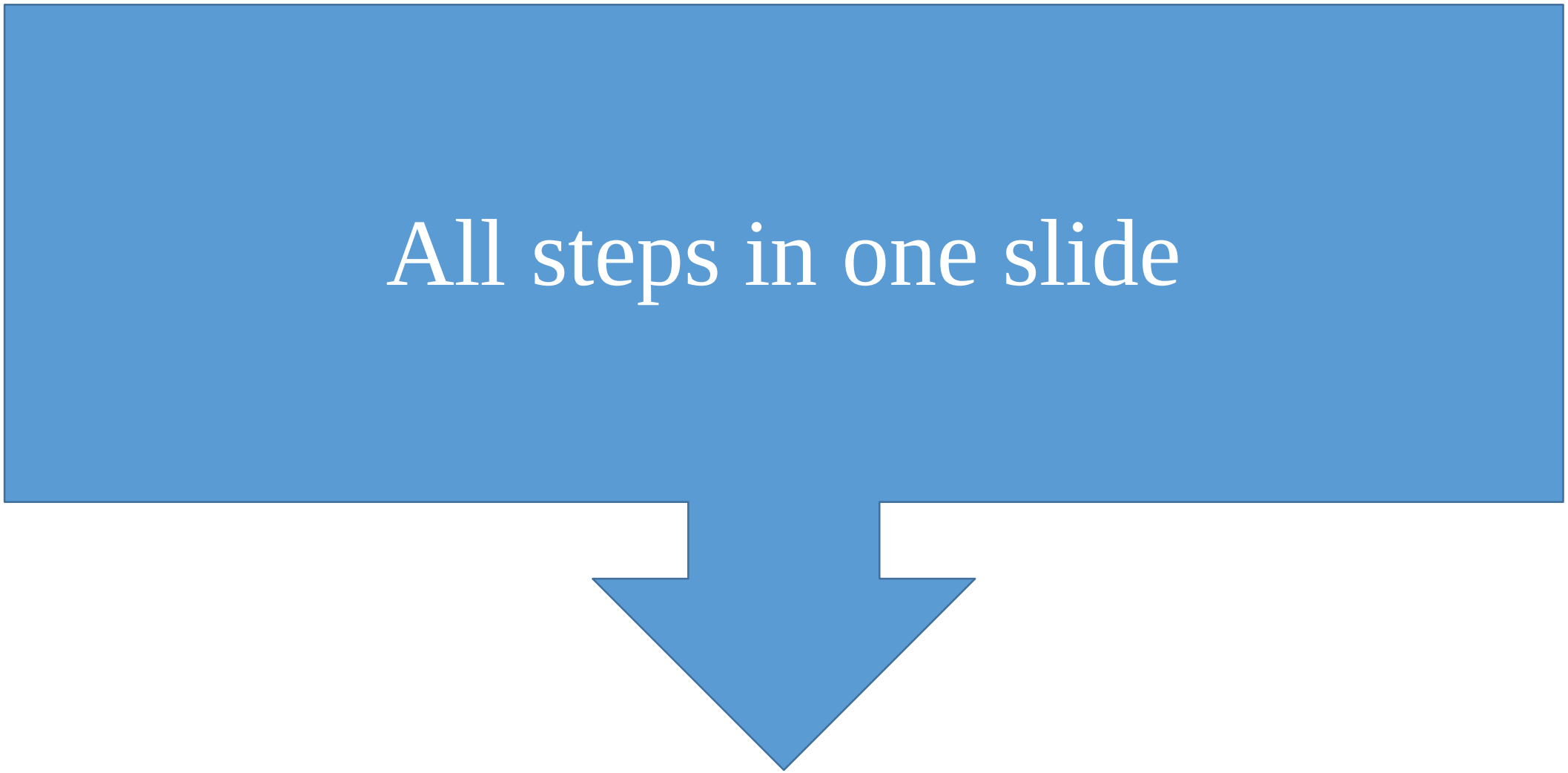


Sodium-Potassium pump

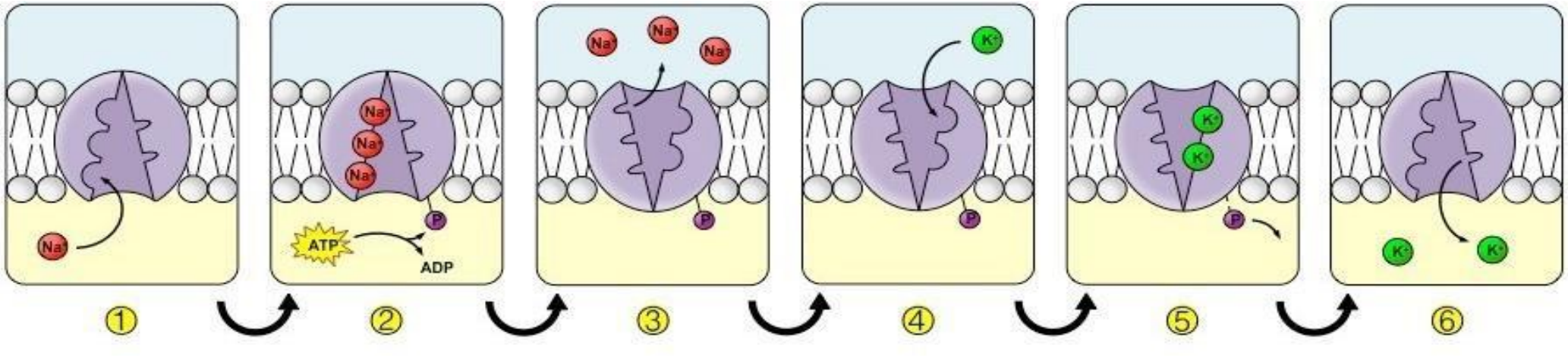
- The two potassium ions enter the cell.

Note: An inward-facing form (inner open) binds highly to sodium, and an outward-facing form (outer open) binds highly to potassium.




All steps in one slide

1. Na/K pump opens to the inside of the cell, and three sodium ions bind to the sodium-potassium pump.
2. Hydrolysis of ATP to ADP and phosphate.
3. Phosphorylation of the pump, then Conformational change of the pump (opens towards the outer side), and exit of the three sodium ions.
4. Binding of two potassium ions to the extracellular site of the pump.
5. The phosphate group is removed from the pump (dephosphorylation).
6. Conformational change of the pump (opens towards the inner side), and the two potassium ions enter the cell.



Secondary active transport

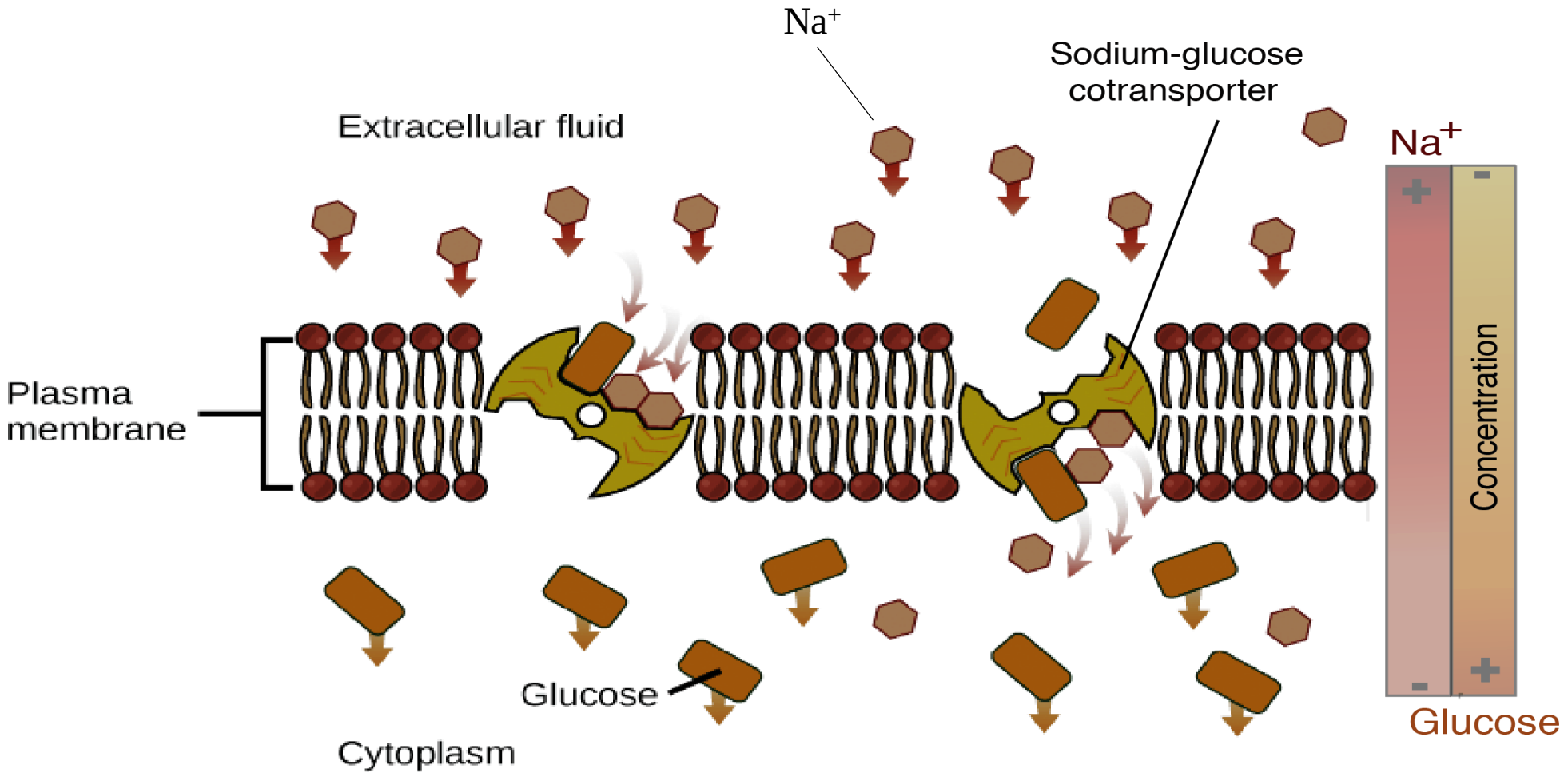
- In this type, energy is not obtained from ATP, but from the electrochemical gradient, which released as the ions move down their gradients.

- **An electrochemical gradient** is the difference in the concentration of ions across the membrane.

- A carrier protein helps to move molecules through cell membrane, with the help of ions (ex. Sodium).

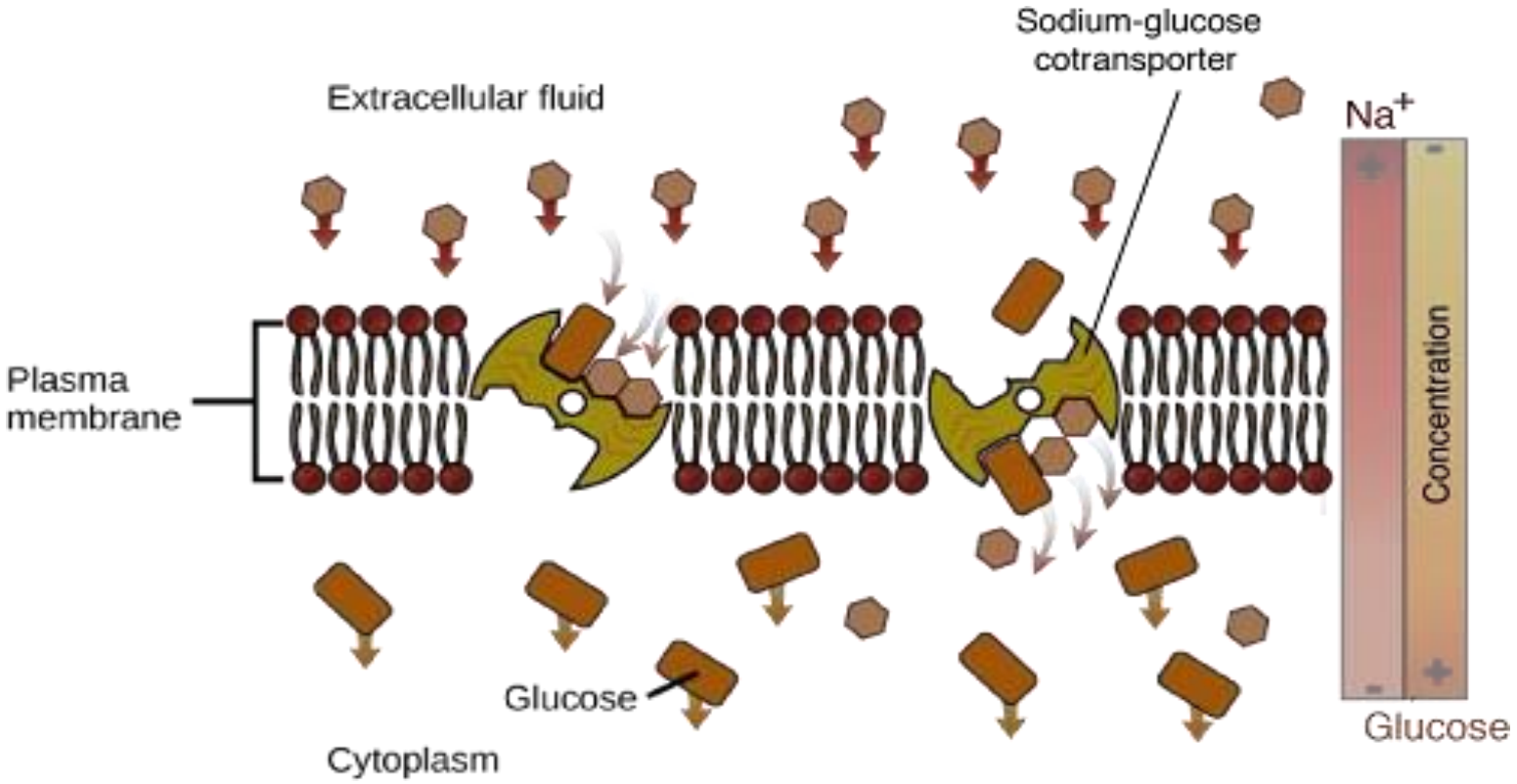
- It is the movement of molecules against their concentration gradient (ex.glucose), with the help of protein transporter and ion (ex. Na^+)

Secondary active transport



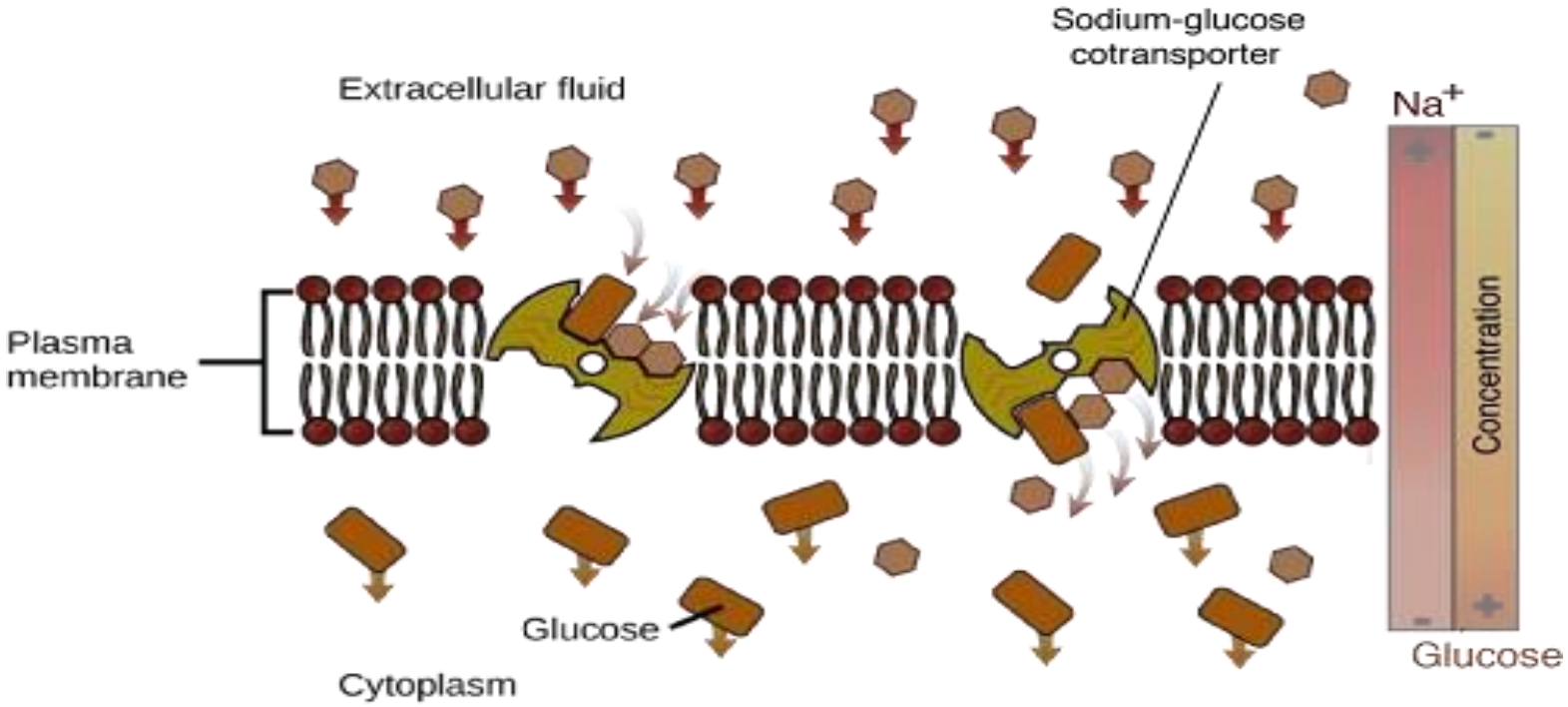
Sodium-glucose transporter

- They transport glucose and found in small intestine.
- The sodium-glucose pump bind to two sodium ions and glucose.



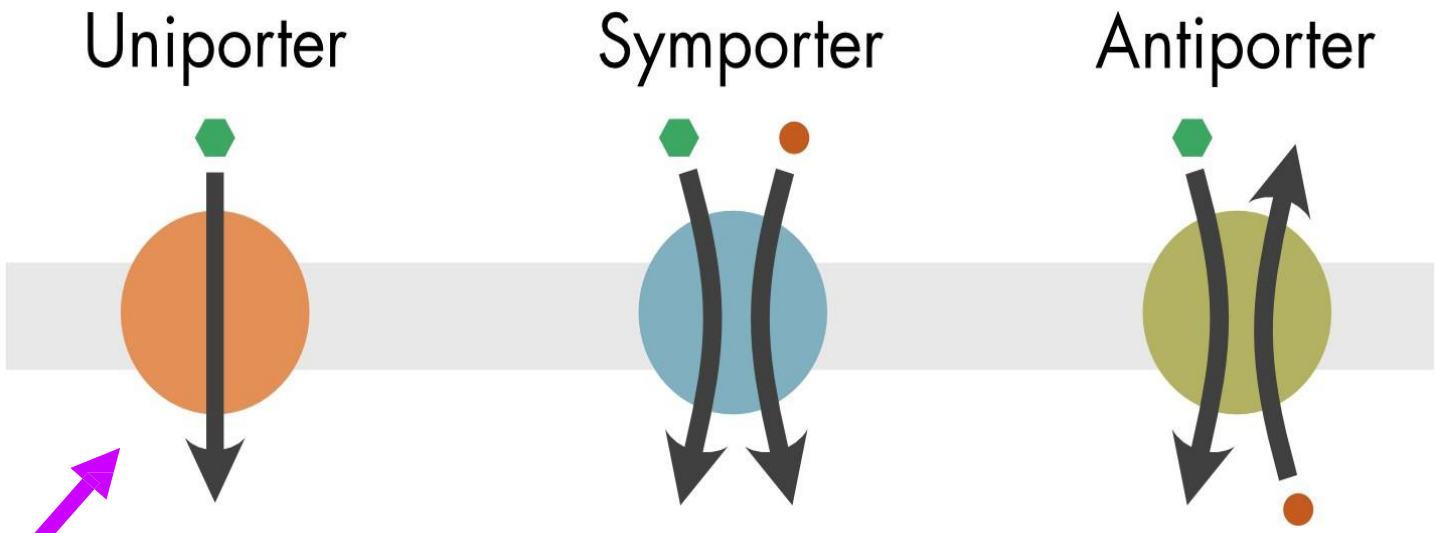
Sodium-glucose transporter

- Sodium enters the cell, and takes the glucose into the cell with it.
- The sodium-glucose pump uses secondary active transport because it combines the transport of sodium down its concentration gradient with the transport of glucose against its concentration gradient.



Types of porters or transporters

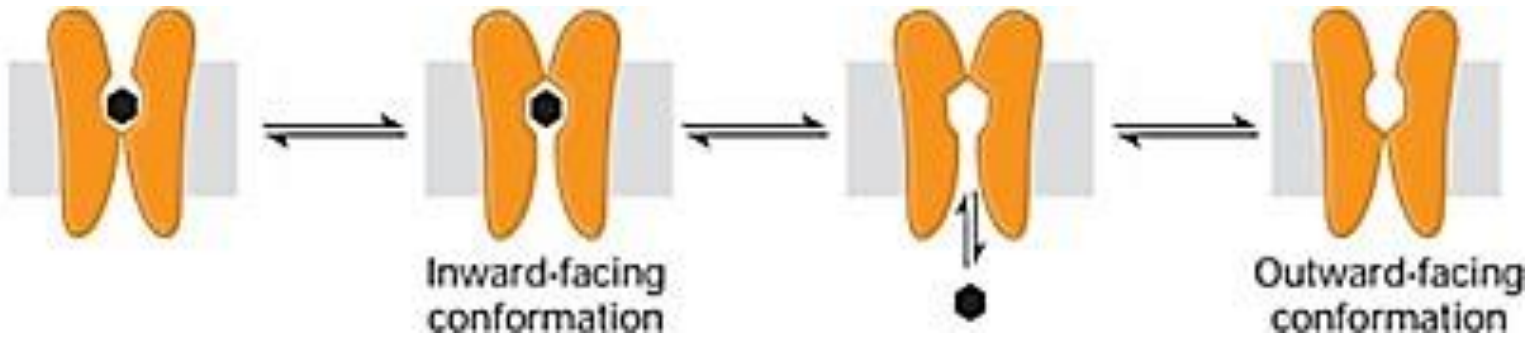
1. Uniporter
2. Symporter
3. Antiporter



In facilitated diffusion

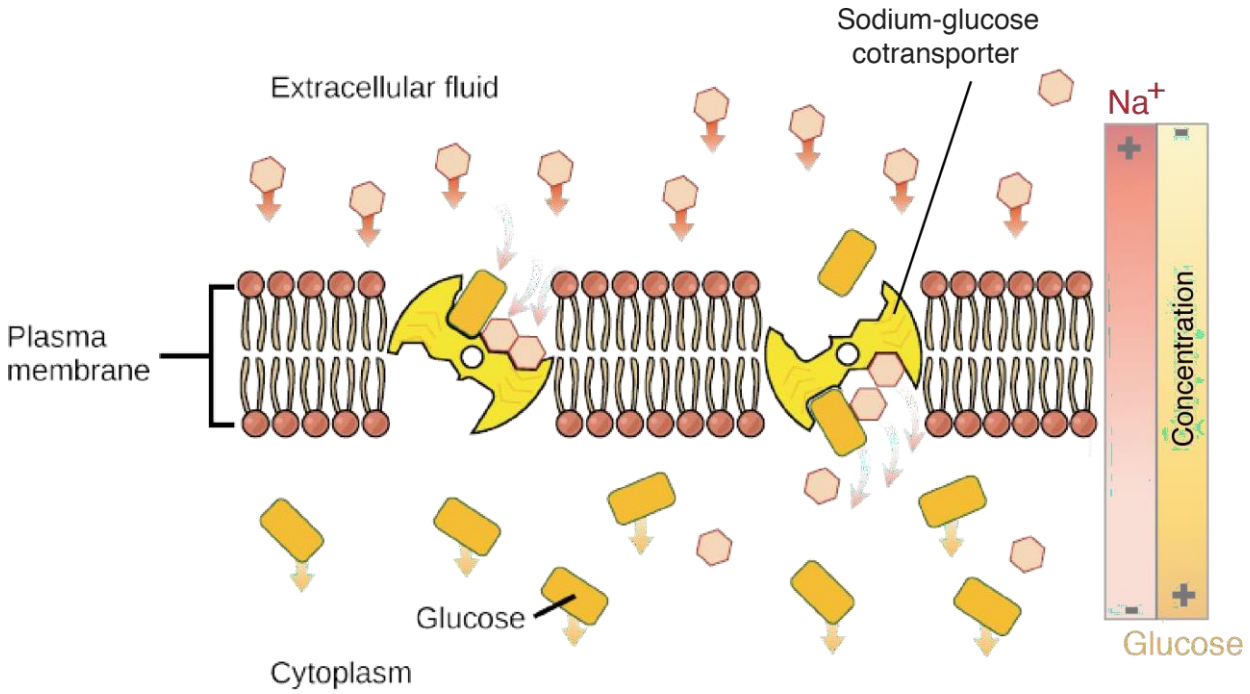
1.Uniporter

- **Uniporter** is a transporter that only moves one molecule across the membrane.
- It is an integral protein and involved in facilitated diffusion.
- It can transport sugars and amino acids.



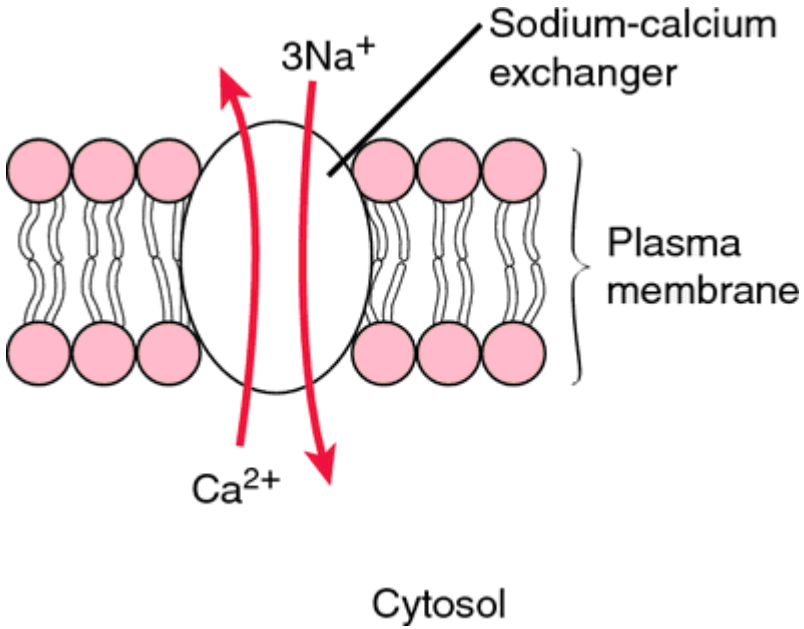
2.Symporter

- Symporter is a transporter that moves two molecules across the membrane in the same direction.
- The uptake of glucose and Na^+ is a symport process (move two molecules in the same direction).



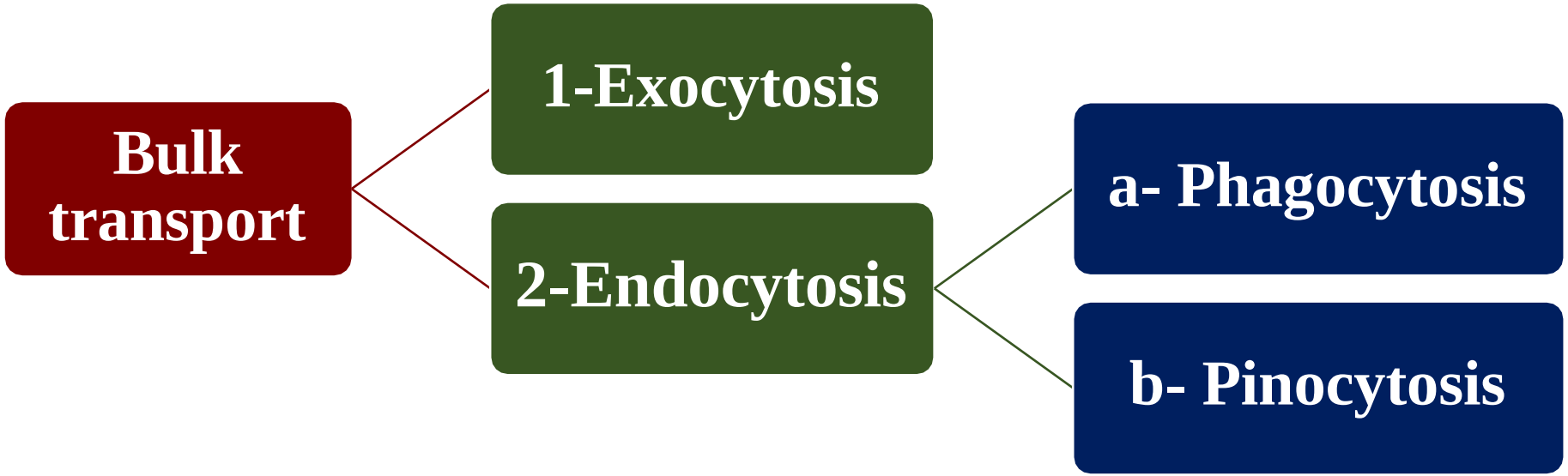
3.Antiporter

- Antiporter is a transporter that moves two molecules across the membrane in opposite directions.
- The example is sodium-calcium antiporter.
- This antiporter enters 3 sodium ions into the cell and exits one calcium ion outside the cell.



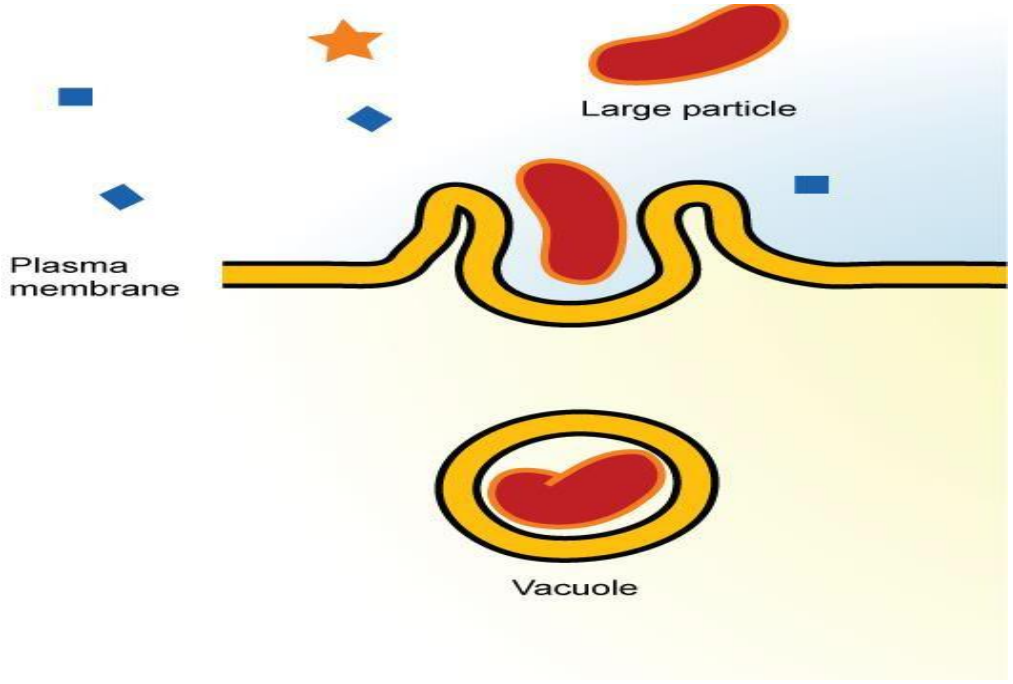
Bulk transport

- It is called bulk transport because the molecules transported are bulk (large molecules).
- Proteins such as carrier and channel proteins can’t be used to transport bulk molecules.
- So, vesicles are responsible for transport.

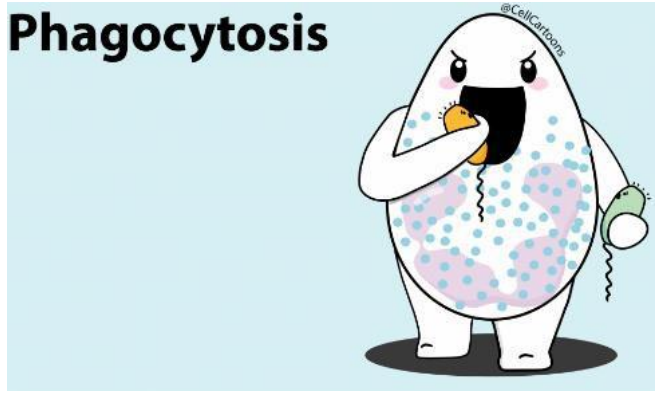


1. Endocytosis

- In endocytosis, large molecules are transported into the cell.
- In endocytosis, the plasma membrane of the cell invaginates (infolds), forming a hole around the target particle.
- After that, the particle is enclosed, and makes an intracellular vesicle.



A- Phagocytosis



- Phagocytosis (means "cell eating").
- It is the process by which large particles, are taken in or swallowed by a cell.

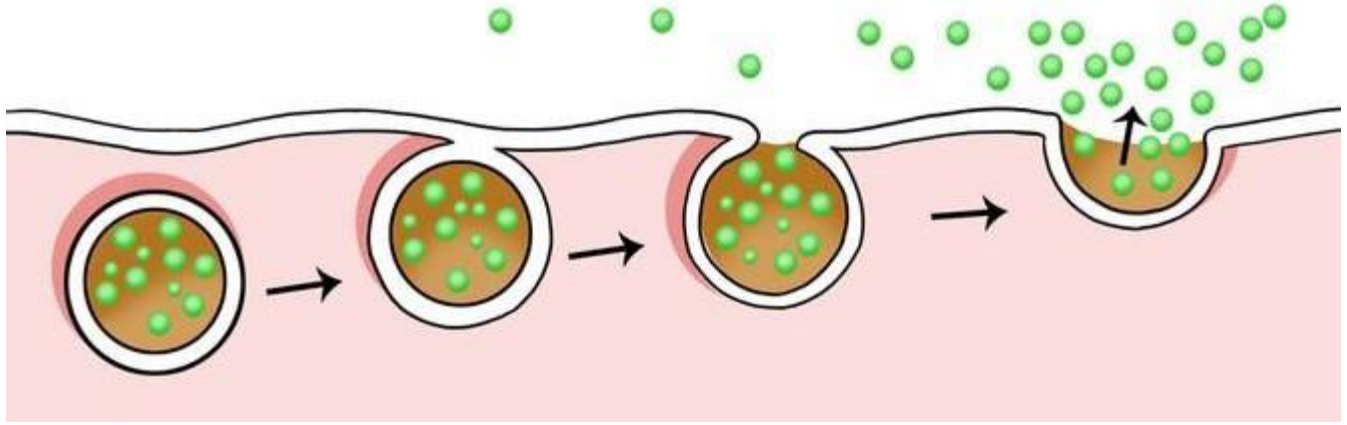
B- Pinocytosis



- Pinocytosis (means "cell drinking").
- This is a process where the cell takes in water and dissolved substances which the cell needs from the extracellular fluid.

Exocytosis

- The process of moving material out of the cell is exocytosis.
- It is important to remove wastes out of the cell.



Questions

I. Complete the following:

1 The sodium-potassium pump transports sodium ……. the cell and potassium……..….the cell in a repeating cycle.

2 The type of sodium-calcium porter is……

3 …….... is the movement of molecules from Lower to higher concentration.

4 ……is a type of transport that moves large molecules into a cell.

5 The type of porter used in facilitated diffusion is……..……

6 The two types of active transport are……..…….and……..……..

7 In……..……..transport, the molecules transported are large in size.

II-Mention the importance of sodium-potassium pump.


Any
questions